

18101 SERIES

1.8 INCH (45 mm) LED 7-SEGMENT DISPLAY COMMON CATHODE

FEATURES

- 1.8 inch (45 mm) character height
- Common-cathode configuration
- Wide viewing angle
- High brightness and uniform segments
- Reliable and robust performance

GENERAL SPECIFICATIONS

PARAMETER	SPECIFICATION / VALUE
Series	18101
Category	LED 7-Segment Display
Mode	Common-Cathode (CC)
Character Height (Size)	1.8 inch (45 mm)
Overall Dimensions (L × W)	56 mm (L) × 38 mm (W)
Forward Current (I_F)	20 mA
Pulse Forward Current (I_{FP})	100 mA
Reverse Voltage (V_R)	5 V ($I_R = 100 \mu A$)

ABSOLUTE MAXIMUM RATINGS

($T_A = 25^\circ C$ unless otherwise specified)

PARAMETER	SYMBOL	MAX. RATING	UNIT
Pulse Forward Current ¹	I_{FP}	100	mA
Continuous Forward Current	I_F	20	mA
Reverse Voltage Segment	V_R	5	V
Operating Temperature Range	T_{OPR}	-25 to +75	$^\circ C$
Storage Temperature Range	T_{STG}	-25 to +85	$^\circ C$

Note:

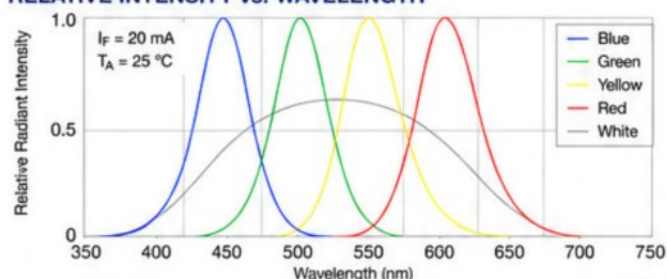
1. Pulse duration ≤ 10 ms with duty cycle $\leq 1\%$.

TYPICAL FORWARD VOLTAGE (V_F) @ 20 mA

($T_A = 25^\circ C$ unless otherwise specified)

LED COLOR	TYPICAL V_F @ 20 mA	MAXIMUM	UNIT
Red	1.8 – 2.2	2.5	V
Orange	2.0 – 2.2	2.5	V
Yellow	2.0 – 2.2	2.5	V
Green (Standard)	2.0 – 2.4	2.6	V
Blue	3.0 – 3.4	3.6	V
White	3.0 – 3.4	3.6	V

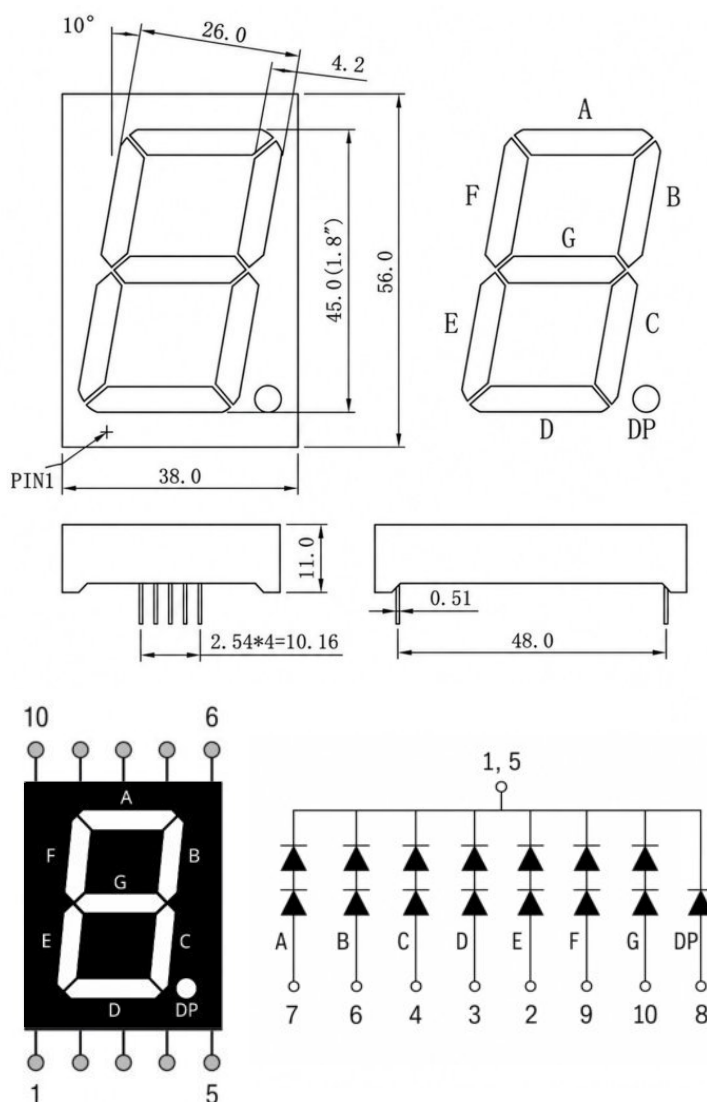
RELATIVE INTENSITY vs. WAVELENGTH



GENERAL DESCRIPTION

The 18101 series is a 1.8 inch (45 mm) LED 7-segment display designed for high visibility applications. The display features uniform brightness, low power consumption, and a robust package suitable for a wide range of electronic devices and industrial equipment.

PINOUT (TOP VIEW)



EMITTED COLOUR / WAVELENGTH

($T_A = 25^\circ C$)

Emitted Colour	Peak Wavelength λ_p (nm)	Spectral Line Half-width $\Delta\lambda$ (nm)
Red	630	20
Yellow	590	35
Blue	470	40
Green	517	40
White	x: 0.27, y: 0.28	—